

Implications of mergers and acquisitions for information disclosures in earnings calls

Roberto Ragozzino^{a,*}, Jeffrey J. Reuer^b

^a Nova School of Business & Economics, Urbanização S. Gonçalo, Rua da Holanda, 1, 2775-405, Carcavelos, Portugal

^b University of Colorado, Leeds School of Business, 995 Regent Drive, Koelbel Building, Campus Box 419, Boulder, CO, 80309, USA

ARTICLE INFO

Keywords:

Corporate strategy
Acquisitions
M&A
Information asymmetry
Adverse selection
Earnings calls
Corporate communication
Executives
Analysts

ABSTRACT

We examine the linkages between firms' M&A activity and the information exchange between executives and analysts during quarterly earnings calls. Prior M&A research has mostly focused on the considerations presented by information asymmetry between buyers and sellers themselves in M&A. However, these concerns also exist between the buying firm's executives and capital markets. We examine whether firms' M&A activity results in an information exchange between executives and analysts that highlights corporate strategy as a discussion topic. Thus, we test whether there is consistency between firms' corporate growth agendas and the information released to the public by these firms. This question is relevant, as disclosure is central to firms' ability to raise capital and compete. We also investigate the degree to which these information exchanges affect analysts' earnings forecast errors.

1. Introduction

Asymmetric information plays a central theoretical role in corporate strategy (e.g., Balakrishnan and Koza, 1993; Vanhaverbeke et al., 2002; Villalonga and McGahan, 2005; Higgins and Rodriguez, 2006; Schildt and Laamanen, 2006; Capron and Shen, 2007; Graebner, 2009; Zaheer et al., 2010; Cuypers et al., 2017; Humphrey-Jenner et al., 2017). Many corporate transactions are characterized by some level of information asymmetry, and this leads a less informed buyer to face adverse selection, or the risk of overpaying for assets (e.g., Akerlof, 1970; Ravenscraft and Scherer, 1987; Healy and Palepu, 1993; Healy and Palepu, 2001). It is also relatively well understood that the hazards associated with information asymmetry tend to be greater when the value of a transaction is significant to a buyer, and when the transaction is a one-off deal, rather than an ongoing relationship that will have buyers and sellers engage in repeat business over time (Riley, 2001). As a consequence, mergers and acquisitions (M&A) are especially prone to adverse selection concerns.

M&A also happen to be a pervasive investment vehicle supporting corporate strategy: According to the specialized analytics and technology company Dealogic, the yearly dollar volume of worldwide activity has ranged between \$2.2 and \$4.1 trillion during the 2015–2019 timeframe.¹ Given the economic importance of M&A and firms' exposure to adverse selection in these deals, academic studies in strategy, finance, and other fields have devoted significant attention to examining the implications of adverse selection for M&A deals and firm performance. The evidence brought forth by this body of work confirms the broad point that adverse selection

* Corresponding author.

E-mail addresses: roberto.ragozzino@novasbe.pt (R. Ragozzino), jeffrey.reuer@colorado.edu (J.J. Reuer).

¹ <https://www.dealogic.com/insight/ma-highlights-full-year-2019/>.

plays a role in explaining firms' M&A strategy and performance (e.g., Fowler and Schmidt, 1989; Villalonga and McGahan, 2005; Moeller et al., 2007; Chemmanur et al., 2009; Officer et al., 2009; Cheng et al., 2016; Cuypers et al., 2017; Martin and Shalev, 2017; Yakis-Douglas et al., 2017; Borochin et al., 2019).

Despite this fairly clear-cut conclusion, it is noteworthy that prior work has focused mostly on the asymmetric information present between the buying and the selling firms in M&A and the implications it presents during M&A dealmaking. This work has applied the theoretical models of information economics originally developed for other markets to the M&A market context, in order to understand the antecedents and consequences of various aspects of M&A transactions. Specifically, M&A studies have relied on seminal research such as Akerlof's (1970) classic model of used car markets, Spence's (1973) model of signaling in labor markets, and Stiglitz's (1973) analysis of insurance markets (e.g., Reuer, 2009). The upshot of research on M&A is that under conditions of information asymmetry, deals may fall through, or alternative entry modes may be used, absent other remedies such as contractual provisions or payment structures to reduce the exposure of buyers to adverse selection risks (e.g., Balakrishnan and Koza, 1993; Kohers and Ang, 2000; Datar et al., 2001; Gilson and Schwartz, 2005; Cain et al., 2011; Reuer and Ragozzino, 2012; Macias and Moeller 2016; McCann et al., 2016).

However, there is a separate adverse selection problem in M&A markets that has often been neglected in previous research (for an exception see Yakis-Douglas et al., 2017): Information asymmetry may also exist between company executives carrying out M&A and the very shareholders they represent. This means that even if a buyer were to address information asymmetry satisfactorily between itself and a seller during deal negotiations, it would still face the critical challenge of relaying its M&A strategy to shareholders. It is noteworthy that such third parties often do not appear in other market settings involving buyers and sellers in which the original models of adverse selection were based (e.g., the original models applied to used cars, labor markets, insurance, etc.). For shareholders, the more complex the transaction is, the higher their information-gathering costs will be, and they obtain diminishing benefits from trying to make sense of the firm's deals (Veldkamp, 2006). As the information becomes costly to gather or process, shareholders can discount the value of the firm and firms will experience a higher cost of obtaining external resources.² It follows that the consequences of information asymmetries in M&A markets might even be more far-reaching than previous M&A studies on this topic have considered, and the costs they entail for acquirers have been understated. This oversight may be in part due to the greater attention that corporate strategy scholars have given to moral hazard concerns – another implication of information asymmetry involving shareholders that can give rise to value destroying acquisitions (e.g., Jensen and Meckling, 1976; Dalton et al.,). Alternatively, scholars may have overlooked adverse selection in this context because they have implicitly assumed that capital markets are informationally efficient (Fama, 1970; Milgrom and Stokey, 1982).³ Regardless, for the most part research in the area of corporate strategy in general, as well as M&A in particular, has neglected the importance of adverse selection between company insiders and shareholders. A recent notable exception is the work by Benner and Zenger (2016), who have provided a conceptual discussion of the challenges managers face when they devise and implement complex strategies and subsequently have to relay information on these strategies to capital market participants.

We wish to build on these ideas and join them with information economics research on M&A. We do so by examining whether M&A carries important implications for information asymmetries in capital markets and their operation. We first set out to investigate whether the information disclosures by company insiders to shareholders follow a pattern that is consistent with the company's strategic complexity, as defined by its M&A activity in particular. To this end, we track companies' information disclosures by using a comprehensive set of earnings calls transcripts spanning 12 years and over 6,000 unique companies and millions of exchanges between executives and analysts. In particular, we are able to isolate the corporate strategy terms spoken by company executives and analysts during earnings calls. Further, in view of potential endogeneity concerns, we adopt a propensity score matching technique to create comparable samples of M&A-active and M&A-inactive firms, as well as of firms engaged in different levels of M&A activity.

Our results demonstrate that the M&A strategies implemented by companies affect their information exchanges with analysts, both in terms of questioning by analysts and the information disclosures by executives. For future research, these findings provide initial evidence that executives engaging in M&A are not only contending with information asymmetries directly in M&A markets with targets during deal-making, but also in other factor markets as well (i.e., capital markets). We therefore conclude that previous M&A research has understated the consequences of the inefficiencies and frictions presented by information asymmetries surrounding acquisition deals. In subsequent analyses, we also show that the magnitude of analysts' earnings forecast errors increases with their doubts regarding a firm's strategy and decreases with the information conveyed by executives during earnings calls. In summary, we offer fine-grained evidence that information disclosure is proportional to the complexity of the firm's M&A strategy, and that richer disclosure by executives and more intense questioning by analysts have a direct impact on the ability of information intermediaries in capital markets to evaluate firms.

As its main contribution, this study provides both conceptual arguments and empirical evidence that the strategic complexity of firms' M&A strategies needs to be articulated to shareholders, in order for the latter to acquire the necessary information to evaluate

² Paradoxically, as noted in recent work, the more unique a firm's corporate strategy, the more difficult it is to communicate it effectively to shareholders, yet it is through a unique and complex corporate strategy that economic rents are possible in the first place due to inimitability (i.e., Litov et al., 2012).

³ The definition of strong form market efficiency is that of a market in which prices reflect all public and private information. Effectively, this definition rules out the problem of adverse selection between company insiders and shareholders, because any deviation in price from the true value of the firm must be random, and not due to a recurring problem of information asymmetry between some firm stakeholders and others. Whether and when assumptions about different levels of market efficiency are warranted is beyond the scope of this paper, and we direct readers interested in this topic to the pertinent work in financial economics (e.g., Grossman and Stiglitz, 1980; Shiller, 1989).

complex transactions by these firms. For the M&A literature, the evidence underscores the point that executives are not only called upon to implement M&A strategies that have the potential of conferring a competitive advantage to their firms – an arduous task in itself given the frequency with which acquirers fail to capture value due to information costs and other challenges – but they are also called upon to articulate the corporate strategy surrounding their investment decisions to the investing public. Our arguments and findings therefore call for future research that bridges economic transactions in certain factor markets (e.g., M&A) with information disclosures in others (e.g., capital markets) to account more fully for the consequences of information frictions affecting corporate strategy.

2. Background theory

The theory of information economics provides important insights into the structure and functioning of exchanges in various markets. The early work of [Akerlof \(1970\)](#) in particular helped lay the theoretical foundations to understand adverse selection, which is a direct consequence of asymmetric information between buyers and sellers. In a nutshell, Akerlof relied on the market for used cars as the context of his analysis and proposed that in a world in which buyers cannot distinguish good and bad cars, buyers either refrain from buying cars altogether – in which case no cars are sold and markets do not clear – or they end up with cars having a low average quality. Research by [Spence \(1973\)](#) and [Stiglitz \(2002\)](#) in other market contexts has further paved the way to help to understand phenomena that could not be fully explained by traditional neoclassical theory ([Löfgren et al., 2002](#)). The applicability of information economics has since been proven to reach far and wide across various market contexts, including banking and credit ([Stiglitz and Weiss, 1981](#)) and financial markets (e.g., [Grossman and Stiglitz, 1980](#); [Myers and Majluf, 1984](#); [Healy and Palepu, 2001](#)), insurance products ([Rothschild and Stiglitz, 1976](#)), labor market contexts ([Shapiro and Stiglitz, 1984](#)), corporate social responsibility ([Grewal et al., 2019](#)), and impression management during CEOs' public presentations ([Whittington et al., 2016](#)), among others.

2.1. Information economics and M&A

Scholarly work has applied the theory of information economics to M&A and for some time has studied its implications for the efficiency of these markets (for a review see [Reuer, 2009](#)). The basic idea, which mirrors the market for used cars discussed by Akerlof, is that sellers in M&A often hold better information on their own firm than buyers. As such, a seller has a natural incentive to inflate its value by presenting the firm's resources and prospects in a favorable light, while a buyer faces due diligence costs and discounts the seller's valuation given its inability to judge the target's unobserved resources and prospects ([Ravenscraft and Scherer, 1987](#)). These ideas have been extensively discussed in early financial economics research, and this work has provided the theoretical basis for identifying ways in which buyers may cope with information asymmetry, whether through information gathering and intermediation, relying on signals, or using protective contractual provisions and payment structures in M&A deals (e.g., [Fishman, 1989](#); [Eckbo et al., 1990](#); [Datar et al., 2001](#)). These early ideas have been subsequently followed by substantial empirical research – including in strategy – which has furthered our understanding of how adverse selection affects buyers and sellers involved in M&A.

While this research has led to considerable insights into M&A activities by firms and their performance implications, this work might still be understating the importance of information costs in M&A markets. This is because this research focuses on the information asymmetry between buyers and sellers during dealmaking, rather than the information asymmetry that can also exist between the buying firm and this firm's own shareholders. To be clear, we are referring specifically to the application of information economics and the adverse selection problem to this context. There is in fact a sizable amount of research that has investigated how information asymmetry affects moral hazard and gives rise to agency problems in public corporations. It is worth drawing a quick distinction between the two problems: Adverse selection concerns itself with the *ex-ante* problems that surface when M&A buyers and sellers hold different information on the target company. As previously discussed, the implications of this predicament are either that buyers will walk away from otherwise viable transactions, or that they will risk overpaying for targets, absent appropriate remedies. In contrast, moral hazard focuses on the problem of hidden actions that emerge from principals (i.e., the shareholders) and agents (i.e., the executives) holding diverging incentives (e.g., [Jensen and Meckling, 1976](#)). In this case, the inability of the principals to monitor agents may lead the latter to engage in large or diversifying acquisitions that might not be in shareholders' interests. This problem, a problem of hidden action, is distinct from adverse selection, a problem of hidden information, though both concerns ultimately find their roots in information asymmetry between different actors and can sometimes be intertwined as a consequence (see [Stiglitz, 2002](#) for a discussion).

It remains for us to explicate why a shift in focus from information asymmetries between buyers and sellers in M&A markets to information asymmetries between those acquirers and their shareholders is warranted and interesting. Generally speaking, it is well known that capital market participants (i.e., investment bank analysts, mutual and hedge fund managers, shareholders, etc.) are called upon to value public firms by using the information made available by these firms' required disclosures. While the valuation of assets in place may be relatively straightforward, it is an altogether different challenge to evaluate strategies that involve forward-looking plans ([Mintzberg, 1987](#)), complex decisions ([Ghemawat, 2005](#); [Gavetti and Rivkin, 2007](#)) and the accumulation and deployment of bundles of resources and capabilities ([Barney, 1986](#); [Montgomery and Wernerfelt, 1988](#); [Zenger et al., 2011](#)). These points have been made by [Litov et al. \(2012\)](#), who noted that it is not only plausible that company insiders possess private information, but it is a requirement for their pursuit of competitive advantage (including via M&A), so that close rivals are not to be able to readily detect this information and be in a position to imitate the firm's strategy.

Company executives leverage such private information when they reach out in factor markets and set out to combine their own resources with those of other companies. At a minimum, M&A result in the additive combination of two previously independent

companies, but their complexity can easily escalate when these deals involve the creation of economies of scale and scope, the redeployability of resources, or other idiosyncratic advantages tied to location, experience or learning. Indeed, such objectives represent the backbone of the value adding proposition of many M&A, and the likelihood of success of these transactions rests on the unique foresight held by company insiders. As noted by Angwin (2007), M&A success may simply mean life or death for organizations. Paradoxically, as the role of private information rises in M&A markets, so does the need for company insiders to relay their motives to capital market participants, in order for the latter to develop informed expectations on the value of the firm (Benner and Zenger, 2016). Should this information be incomplete or missing, investors would face adverse selection and its associated inefficiencies, so tensions can naturally arise between the firm's pursuit of competitive advantage and its need to disclose information to capital markets to obtain funds on attractive terms.

It is also worthwhile noting that a relatively new stream of research has expounded the merits of a so-called "open strategy practice" (Whittington et al., 2011), which includes a number of steps such as blogging, jamming, knowledge-sharing, crowdsourcing and more generally open communication within and across organizations and stakeholders. For example, in a recent paper Yakis-Douglas et al. (2017) examine information disclosure in the area of M&A and surrounding deal announcements, finding that voluntary discussions of individual transactions by companies tend to result in value creation for firms.

In sum, information asymmetry between company insiders and capital markets in M&A is a compelling issue for two main reasons: First, at a basic level, M&A buyers themselves face challenges in assessing the value of target firms. To the extent that buyers move forward with a transaction, they will have ostensibly overcome these challenges satisfactorily. However, even if this is the case, the resolution of the information asymmetry problem separating buyers from sellers still requires buyers to be able to convey information on the transaction to capital markets (e.g., the value creation logic for the transaction, the deal structure, execution plans, etc.). Second, company insiders are also called upon to explicate their strategic vision as to how the M&A transaction will augment the competitiveness of their firms and support their corporate strategy. Here again, the more nuanced the foresight of insiders, the more likely the M&A will yield unique value, but also the more difficult the deal will be to convey to capital markets and might be imitated by others. These two sources of adverse selection are distinct but not mutually exclusive, and they also carry important implications for how firms may be able to raise capital and be competitive.

2.2. The role of earnings calls for information delivery to capital markets

The U.S. Securities and Exchange Commission (SEC) imposes information disclosure requirements on publicly held companies trading on U.S. stock exchanges. For example, firms are required by the SEC to report audited financials each quarter, and this information plays a central role in influencing shareholders' future expectations of companies and hence valuations. In addition to the mandated financial performance disclosures, capital market participants also rely on voluntary quarterly earnings conference calls. These conference calls provide stakeholders with opportunities to engage with firms' corporate executives and to ask spontaneous questions in order to understand better the company's strategy and performance (Lee, 2016). Accordingly, research in disciplines ranging from accounting to financial economics and management has sought to examine how the information disclosed during the calls can affect the firm and its stakeholders (e.g., Frankel et al., 1999; Bushee et al., 2004; Kimbrough, 2005; König et al., 2018; Pan et al., 2018; Harrison et al., 2019; Graf-Vlachy et al., forthcoming).

As a few illustrations, some scholars have examined the roles of analyst coverage (Matsumoto et al., 2011) and information asymmetries (Bushee et al., 2018) on the content of information covered in earnings calls, while others have tied executives' tone or their affective state to ensuing analysts' stock recommendations (Mayew and Venkatachalam, 2012; Henry and Leone, 2005). There is a vast and growing body of work that has examined earnings calls using many different theoretical lenses across these disciplines (for a review, see Gao et al., 2016). Information disclosures might occur by other means, but this recent research establishes that earnings calls are a critical platform used by company insiders to relay information to capital market participants and, in turn, for capital market participants to perform due diligence as well as try to abate information asymmetries between capital providers and insiders. On this basis, below we specify our formal predictions for empirical analysis.

3. Research hypotheses

As discussed, earnings calls provide a unique opportunity for information discovery. Although earnings calls are discretionary and can have downsides such as a risk of information being disseminated in product markets, firms have an incentive to hold them for at least two broad reasons: First, not doing so would likely indicate an intent to withhold information from markets, with potentially adverse valuation consequences. Second, earnings calls provide executives with an opportunity to explain the drivers of their firms' performance, as well as to lay out their strategy and future pursuits. This purpose is beneficial to firms because it allows them to access capital markets, which is the main advantage of being publicly traded, but which requires transparency to providers of financial resources (Chemmanur and Fulghieri, 1999). It is therefore no surprise that a recent survey conducted by the National Investor Relations Institute reports that 97 percent of all public companies hold quarterly earnings calls on a regular basis (<https://www.niri.org/>).

4. M&A as an antecedent of corporate strategy information disclosures

Besides the general incentive to release information through earnings calls, there is also evidence that this incentive will grow proportionally with the complexity of the firm, for instance, based on extraordinary items to be covered by executives or details on how

the firm will be achieving growth (e.g., Frankel et al., 1999). Specifically, single-business firms having relatively predictable performance will likely feature short and basic questions from analysts, as well as cursory responses and explanations by executives, as information asymmetries are likely to be relatively small, all else being equal. By contrast, more complex strategies made up of a multitude of corporate transactions and diverse areas of growth can lead to extended earnings calls featuring numerous and involved questions that enable deeper probing of specific topics.

M&A are among the most resource-intensive strategic investments firms employ, as they require significant capital commitments, costly due diligence, and challenging integration processes. It is partly because of the complexity of M&A deal-making and execution that equity markets often respond skeptically to M&A announcements (e.g., Kaplan and Weisbach, 1992), as also corroborated by a recent meta-analysis showing a consistent slightly negative reaction to acquisition announcements on average (i.e., King et al., 2021). It follows that analysts will also want to leverage earnings calls as opportunities to obtain otherwise opaque information on the objectives held by executives and how they will be carried out. Thus, we expect that analysts will actively inquire about these deals and raise broader concerns about corporate strategy to executives during calls. Likewise, based on their need to preserve liquid capital markets for their firms in order to obtain external financing on attractive terms, executives will have to provide pertinent information on their M&A, because failure to disclose this information can jeopardize their firm's access to capital markets. As evidence of this, research in accounting has argued that executives use conference calls to convey private information surrounding M&A announcements, and this process results in more positive reactions by shareholders (i.e., Kimbrough and Louis, 2011). In sum, we predict a positive relationship between a company's M&A activity and the amount of information exchanged on this activity during an earnings call:

Hypothesis 1. Analysts in earnings calls will inquire more on the firm's corporate strategy, the greater the firm's M&A activity.

Hypothesis 2. Executives in earnings calls will provide more information on the firm's corporate strategy, the greater the firm's M&A activity.

While the above hypotheses might appear intuitive at first glance, there are many reasons why M&A activity might not translate into more meaningful discussions of corporate strategy and that make these hypotheses falsifiable. For example, there is an opportunity cost in covering one topic over another in earnings calls, and inevitably priority must be given to those topics that may be providing shareholders with the most useful information on the firm. Additionally, executives (and analysts) may choose to discuss topics other than the firm's corporate strategy to avoid transmitting information to competitors about their firm's strategy. Likewise, there may also be managerial and opportunistic reasons why executives might not want to engage in discussion of corporate strategy (for instance, if an acquisition is brought about by agency considerations, an executive has an interest in not drawing attention to it). Lastly, there are many other vehicles by which firms may pursue diversification (like joint ventures, licensing, corporate venture capital, etc.). Thus, the use of the corporate strategy dictionary we create is not inherently limited to M&A. Finally, it is not clear that analysts will prioritize plumbing the depths of firms' corporate strategies, given their immediate focus on predicting near-term earnings as accurately as possible.

4.1. Consequences of corporate strategy information disclosures

While the previous section concerned itself with the congruity of firms' M&A strategy and the information exchange in earnings calls, we now turn our attention to the effects of this exchange on the adverse selection problem faced by capital market participants. Namely, we examine whether *ceteris paribus*, analysts' earnings forecast errors in the quarter following the focal call are affected by the information flow during the call. Thus, the previous hypotheses established the antecedent logical links between adverse selection and the mechanisms available to market participants to reduce its effects (i.e., questions by analysts and disclosures by executives). In contrast, the ensuing section attempts to determine whether the mechanisms above facilitate reductions in information asymmetry using analysts' forecast errors as an established indicator of these challenges faced by analysts (Lang and Lundholm, 1996; Coller and Yohn, 1997; Easterwood and Nutt, 1999; Affleck-Graves et al., 2001; Kim, 2016).

More specifically, analysts are tasked with attending calls and raising questions on the aspects of a firm's performance and direction about which they believe they are insufficiently informed. They subsequently use the newly acquired information to form an estimate of what the firm's earnings will be in the quarter following the call (e.g., Bowen et al., 2002; Kimbrough, 2005; Lobo et al., 2017; Heinrichs et al., 2019). It is therefore reasonable to infer that, all else equal, the greater the asymmetry in information between analysts and executives, the more the former will question the latter. Thus, the amount of questioning on a firm's M&A strategy should reflect the level of information available to analysts on this topic. Barring external and perverse dynamics,⁴ if few or no questions are raised, then analysts should hold equal information to executives, and analysts can form forecasts based on information at hand. In contrast, extended inquiries by analysts should indicate a wide information gap between the two, which in turn raises the stakes of adverse

⁴ Some research and anecdotal evidence have noted that the ability by analysts to ask questions may be partially driven by their previous record of making positive recommendations on the firm issuing the earnings call (see for instance Siconolfi, 1995; Chen and Matsumoto, 2006; Mayew, 2008).

selection.⁵ As such, we therefore posit:

Hypothesis 3. The greater the amount of questioning by analysts on the firm's corporate strategy in a given earnings calls quarter, the larger the analysts' earnings forecast error in the following quarter.

Questions by analysts reflect their concerns and information asymmetry regarding M&A within firms' corporate strategies, while answers by executives should have the effect of reducing adverse selection problems for firms whose information is more difficult to understand (e.g., Frankel et al., 1999; Bushee et al., 2018; Rabier, 2017; Bochkay et al., 2020). Although it is likely that there will be a positive correlation between the questions on a firm's M&A activity and the corresponding answers offered by executives, it is also true that executives have considerable discretion of providing differing levels of details on any given question as well as disclosing information on an unprompted basis. In some cases, they may choose to talk at length on transactions' motives and intended objectives, even if they are not questioned by analysts, while in others they may opt to be short and concise, even when pushed by analysts. Likewise, executives can also steer the conversation to one topic or another, based on their perception of what items may deserve more coverage, independent of whether they are asked about them directly or not.

At the most basic level, executives are called upon to allocate the allotted time to covering several topics, of which a firm's M&A agenda is but one. Thus, extensive discussions on any given items may simply not be feasible based on time limitations alone. Moreover, there has been considerable academic work that has examined the drivers of the quantity and quality of the information executives disclose during earnings calls. Two clear-cut takeaways from this research are that there is a positive relationship between the opacity of a firm's performance and strategy and the amount of information conveyed by executives during calls (Frankel et al., 1999), as well as between the amount of information disclosed by executives in earnings calls and the reaction of market participants (see for instance, Kimbrough and Louis, 2011). Even still, both anecdotal and scientific evidence points to a great deal of variability in the information volunteered by executives, such that executives may willingly or unwillingly fail to convey all the relevant information, either because of their own limitations, pre-conceived prejudices, or their desire to protect the firm's strategic purposes, as discussed in prior work (see for instance Francis and Philbrick, 1993; Hollander et al., 2010; Matsumoto et al., 2011; Mayew et al., 2013). Insofar as firms with greater M&A activity offer significantly more information on their corporate strategy, we predict that this information will help to reduce the effects of adverse selection affecting analysts:

Hypothesis 4. The greater the amount of information disclosed by executives on the firm's corporate strategy in a given earnings calls quarter, the smaller the analysts' earnings forecast error in the following quarter.

5. Methods

5.1. Data and sample

We draw from a number of data sources to compile the data required for our analysis. As a starting point, we use quarterly earnings calls transcripts made available through Seeking Alpha, which is an organization that provides investors with news, information, and data on a large set of public firms. In detail, the population of earnings calls available through Seeking Alpha as of the time of our data collection comprises over 103,000 transcripts released by 4,500 unique companies over the 2006–2017 timeframe.⁶ In our sample we only focus on firms traded either on the NYSE or Nasdaq. This screening criterion allows us to avoid smaller firms traded in minor exchanges or "over the counter", which are also unlikely to be covered by Seeking Alpha. If we compare the total market capitalization of the firms in our sample against the total market capitalization of the firms traded on the NYSE and Nasdaq, we find that our sample comprises on average 70 percent of the value of these firms.⁷

The transcripts we use are available to download as text files and they contain every single interaction among company executives, analysts and the operator. Notably, each comment by the individuals is preceded by the identifying information of the speaker, which allows us to isolate with precision who said what in each call. We then use Python to code each of the comments and build a database that uniquely identifies the company issuing the earnings call, the time of the call, the identity of the speaker and the position of the speaker comment in the context of the call. In all, the sample includes 7.27 million individual comments in total.

Having identified the baseline sample of firms for which we have earnings calls' transcripts available, we complement this dataset with financial information from Compustat quarterly data, and with information on earnings forecasts from I/B/E/S. In both cases, we use the same year and quarter as the earnings call to match all databases. The next step is to calculate the M&A activity of the firms in

⁵ It is worthwhile to note that although it is possible that prolonged questioning by analysts will result in a reduction of the information gap separating them from executives, this is not likely after controlling for executives' answers. Also, earnings calls have limited scope and duration, and a firm's corporate strategy is but one of several topics that may need to be covered in a call. As a result, M&A will inevitably only receive a fraction of the total call time, and rarely do analysts manage to ask all the questions they wish, and the questions they do ask may trigger additional concerns about a transaction from other analysts.

⁶ Later in the paper we explore the effects of removing the years surrounding the financial crisis from the analyses.

⁷ Seeking Alpha chooses what firms to cover based on the preferences of its subscribers. A comparison of the average (median) size of our sampled firms against the population of NYSE and Nasdaq firm size reveals significant differences, with the former being about 1.5 (2.3) times larger than the latter. Thus, despite the comprehensive coverage of Seeking Alpha in terms of total market cap, and even though by only considering NYSE and Nasdaq firms we are sampling on the largest public firms in the United States, there remains a size bias, which does not affect our empirical approach, but might affect the generalizability of our findings.

our sample. While M&A is an extremely pervasive tool in corporate strategy, there is a great deal of heterogeneity around whether and to what extent individual firms employ acquisitions. To arrive at a detailed account of just how much each individual firm implemented M&A prior to an earnings call, we draw from the Thomson One database, which is a comprehensive source that canvasses firms' corporate activity over time.

Propensity score matching and endogeneity. The goal of our analysis is to determine whether firms that engage in M&A activity provide useful information to investors, in order to allow the latter to be less exposed to the risk of adverse selection. We are also interested in determining whether analysts who participate in the calls on behalf of capital market participants, inquire about the corporate strategies of firms that engage in M&A. To this end, it is necessary to compare samples of firms with and without M&A activity (as well as compare firms with less versus substantial M&A activity, as we will discuss later). It is apparent that simply bifurcating the population of firms based on whether they engage in M&A creates potential problems, insofar as this sort of activity is financially and managerially demanding, and it is also not suited for all competitive contexts and time periods. Accordingly, such direct comparisons may lead to biased inferences due to unobserved heterogeneity.

To address this concern, we first estimate models predicting the use of M&A by individual firms and create propensity scores that we use to match M&A active firms with their most comparable counterparts. We then build a final sample based on the following criteria: First, the absolute value of the difference between the propensity scores of active and inactive M&A firms has to be smaller than or equal to 0.1. Second, both firms have to be in the same industry⁸ and the earnings call has to be in the same year and quarter. Lastly, both firms have to have the same status as either being in the S&P 500 index or not.⁹ If all of these criteria are met, then the inactive firm is included in the sample of firms to compare against its corresponding active firms. We allow for as many as three matches to a given active firm in a given quarter, though our results are not sensitive to this choice of creating comparison groups. The final sample comprises 55,928 firm-quarters, of which roughly 26 percent are active in M&A. In subsequent analyses, we repeat the same matching process, but we compare only highly active versus less active firms. In this case, we exclude all firms without at least one M&A transaction, and we split the sample between those firms with levels of activity above or below the median level. This supplementary dataset features 20,990 firm-quarters, of which about 5,600 are highly active firms, and the remainder are the matched counterparts.

For the estimation of the models predicting analysts' earnings forecast errors, we begin from the two samples described above (i.e., active vs. inactive firms and highly-active vs. less-active firms) and then implement additional matching procedures, as follows: We isolate firms that feature a corporate strategy dictionary use at or above the median (either on the analyst or the executive side, depending on the model specification) and then match these firms against the pool of firms with below-the-median dictionary usage, based on the propensity scores obtained from the estimation of a logistic regression predicting a high use of the corporate strategy dictionary¹⁰, as well as the same matching criteria discussed previously. In sum, we start from samples of firms that are identical in every respect considered, except for their use of M&A and then use propensity-score matching to compare these firms based on the use of the words contained in our corporate strategy dictionary.

5.2. Variables and specifications

Propensity scores and matching. We begin our analyses by estimating two logistic regressions predicting whether or not the focal firm engaged in M&A (or was heavily engaged in M&A or not), in order to match firms. To this end, we create two indicator variables labeled *M&A active firm*, which function as dependent variables in the models. The first variable takes a value of one if the firm entered into at least one M&A deal in the 5 years preceding the earnings call, and zero otherwise. The second, which only includes M&A-active firms during the same time window, separates firms with an activity level equal to or greater than the median level of activity from firms with an activity level lower than the median. The covariates in the propensity score models mostly overlap with the specification of the models used to test our predictions with one single exception. We discuss these variables in detail below.

Main models. The second set of dependent variables are implemented in the estimation of the models for testing hypotheses 1 and 2, in which we predict the use of corporate strategy words by both analysts and executives during earnings calls. In order to capture the extent to which executives and analysts engage in a conversation surrounding the M&A activity of the firm, we build a dictionary of words that mirrors Feldman's (2020) list of corporate strategy keywords. These words and their variants are: ACQUI*, ALLY, ALLIANCE, ASSET SALE, CORPORATE, DIVERSIF*, DIVEST*, FIRM BOUNDAR*, M&A*, MERG*, SCALE, SCOPE, SELLOFF, SPINOFF, STRATEGY.¹¹ The computation of the variables involves three steps. First, we sum up the dictionary keywords spoken by each of these

⁸ We use the Fama-French 12-industry definition to distinguish the following industries: (1) Food, Tobacco, Textiles, Apparel, Leather, Toys; (2) Cars, TVs, Furniture, Household Appliances; (3) Machinery, Trucks, Planes, Office Furniture, Paper, Computer Printing; (4) Oil, Gas, and Coal Extraction and Products; (5) Chemicals and Allied Products; (6) Computers, Software, and Electronic Equipment; (7) Telephone and Television Transmission; (8) Utilities; (9) Wholesale, Retail, and Some Services (Laundries, Repair Shops); (10) Healthcare, Medical Equipment, and Drugs; (11) Finance; (12) Other – Mines, Construction, Building Materials, Transportation, Hotels, Business Services, Entertainment.

⁹ We use the Compustat Index Constituencies database to determine which firms belonged in the S&P 500.

¹⁰ Note that this approach results in the estimation of four separate regressions. The results of these regressions are not shown, but they are available from the authors. The covariates in the models are: (1) whether the firm belongs in the S&P500 index; (2) the total number of alliances and joint ventures entered into by the firm in the 5 years leading up to the call; (3) the total number of acquisitions entered into by the firm in the 5 years leading up to the call; (4) the firm's financial leverage; (5) the firm's market capitalization; (6) the firm's profit margin; (7) industry controls.

¹¹ We run several robustness checks around the dictionary word list, and we discuss these analyses later in the paper.

two classes of participants within each call transcript. Second, we divide these sums by the total words they spoke during the call. Lastly, we multiply these proportions by 1,000 for scaling purposes. These operations allow us to arrive at a continuous, weighted score that is bounded at zero on the lower end (when no instance of any of the keyword exists in the entire call), and at 1,000 on the higher end (in the hypothetical scenario in which every word spoken belongs in the keyword list).

The final dependent variable aims at measuring the accuracy of analysts' earnings forecasts. We compute this measure of analyst forecast errors as the percentage difference between the earnings estimate produced by analysts following the firm and the actual earnings announced by the latter (Collins and Hopwood, 1980; Cen et al., 2013; Harris and Wang, 2013). Specifically, using I/B/E/S data, we obtain the median earnings estimate for the focal quarter and compute the absolute difference between this number and the actual earnings. We then divide the difference by the median estimate.¹²

Controls. The main theoretical variables in the models are the indicator variables that separate M&A-active firms from their counterparts. We have already discussed the way in which we construct these measures in the previous section of the paper. We also introduce several controls, and we discuss each of them here in the order they appear in the tables. *Firm in the S&P 500 index* is an indicator variable taking a value of one if the firm belonged in the S&P 500 index at the time of observation, and zero otherwise. Firms in the S&P 500 are subject to greater scrutiny by the investing public (Shleifer, 1986) and this may affect their corporate communications. *Number of analysts* is included in our logistic models predicting the likelihood of M&A engagement and it reflects the number of analysts following the firm and forecasting its quarterly earnings, as provided by I/B/E/S. Firms with attractive growth prospects or prominence are likely to attract greater coverage by analysts.

At the firm level, we also include *Firm leverage*, *Firm market capitalization* and *Firm profit margin* in all models,¹³ because these measures of the firm's financial health might affect their likelihood to engage in M&A, the Q&A session in earnings calls, and the ability by analysts to forecast the firm's earnings with accuracy in the future. We use Compustat quarterly data to compute these variables. We also include *Firm Tobin's Q* in the logistic models, as the firm growth prospects as proxied by this ratio may also affect its propensity to carry out M&A deals. Tobin's Q is approximated by dividing the firm's market capitalization by its total assets, as commonly done in the literature (Chung and Pruitt, 1994).

The models estimating analysts' earnings forecast accuracy introduce additional covariates as follows: *Corporate dictionary intensive* is the key theoretical variable in the model, and it is computed in exactly the same way as the dependent variables used for the models predicting the utilization of corporate keywords during earnings calls. The control set includes the same variables previously covered, plus *Transcript length*, which is split between analysts and executives, and it consists of the number of words spoken by either class of participant during the focal earnings call. We expect that the accuracy of analysts forecast will be at least partially affected by the length of the information exchange occurring during the call, independent of the use of the corporate strategy keywords we track. *Total M&A deals* is a count variable tracking the number of deals entered into by the focal firm during the 5 years leading up to the call. We include this control because it is plausible that the earnings of active firms may be more difficult to predict than the earnings of less active or inactive firms. Lastly, we include industry fixed effects in all models. Time fixed effects were never significant in any model either aggregately or even individually, so we exclude them from all estimations for the sake of parsimony.

Models. The models used for the analyses pertaining to the utilization of the corporate dictionary keywords by analysts and executives are Tobit regressions. As previously noted, the dependent variables in these models are continuous and bounded at zero, as we scale the count of keywords by the total number of words spoken by each speaker category. Thus, standard OLS models would be inadequate as they may result in the inappropriate prediction of values less than zero. It should be noted that we also re-estimate the Tobit regressions using fractional regressions, and the interpretation and significance of the results remains unchanged. Lastly, the models estimated to predict the determinants of analysts' forecast errors are standard OLS regressions with robust standard errors, as the dependent variable is the percent deviation of the predicted earnings from the actual earnings, in the quarter following the focal earnings call.

6. Results

The first analytical step in our analysis was to estimate a logistic regression to build the propensity scores used for matching M&A-active and M&A-inactive firms. Specifically, we predicted M&A activity for the broad sample of firms, as well as only for firms with at least one M&A transaction. The dependent variables of these models split firms' level of M&A activities between whether they were at or above the median level of active firms, or below. Further, the covariates isolate firms that belonged in the S&P 500 index, the number of analysts present in the call, the firm's leverage ratio, market capitalization, Tobin's Q and profit margins, as well as industry fixed effects. While we do not report the results of these estimations for brevity, both models were very significant on an overall basis, and nearly all covariates appear to be strongly associated with the dependent variable.

It is useful to offer a visual representation of how well the procedures we followed match firms based on their M&A activity. In Fig. 1a and Fig. 1b we plot the average value of the six covariates in the two models using for the estimation of the propensity scores (i. e., the M&A-active firms) and the matched samples across these groups. The grey area in the plots shows a range of 5 percent above or

¹² It is apparent from the way in which we operationalize this variable that we are interested in earnings forecast deviations in any direction. Thus, we are not predicting positive or negative differences based on the use of the corporate dictionary keywords considered, but rather greater accuracy.

¹³ Leverage is computed as the total debt-to-total assets ratio. Market capitalization is the straightforward multiplication of the firm's share value at the end of the quarter nearest to the earnings call by the number of its outstanding shares. Profit margin is the ratio of the firm's net income to its total sales.

below the M&A-active firms' values. It is evident that the vast majority of matching firms are well within this 5 percent error margin. Thus, we are able to arrive at a closely comparable set of firms independent of their M&A activity.

Table 1 shows basic descriptive statistics and correlations for the full matched sample.¹⁴ It is noteworthy that the use of the corporate dictionary keywords appears to be far more frequent for executives than for analysts, although as expected both classes of call participants appear to use these words concomitantly and when the firm is active in M&A (i.e., all $p < 0.001$). While only roughly 7 percent of the firms in our sample belong in the S&P 500 index, these firms are more active in M&A and they tend to be bigger and more profitable than their counterparts. Moreover, M&A activity is positively correlated with firm size, as expected.

We now turn to the models predicting the use of the corporate dictionary keywords in earnings calls. Our first hypothesis predicted a higher incidence of analyst questions surrounding the firm's corporate activity, when the firm engaged in M&A. We find strong support for the hypothesis, as the *Active M&A firm* coefficient is positive and highly significant. Similarly, when we replace the indicator variable with the count of total M&A deals entered into by the firm in the 5 years preceding the call, we find that each additional transaction positively affects the analysts' questioning of the firm's corporate strategy (i.e., both $p < 0.0001$). The difference between M&A active and inactive firms amounts to a nearly 9 percent greater usage of corporate strategy terms by analysts.

The same pattern is also observed when we examine the executives' use of the corporate dictionary keywords: Both the dummy variable separating active and inactive firms, and the continuous deal count variable are positive, sizable and highly significant (i.e., both $p < 0.0001$). Paralleling the interpretation offered above, M&A active firms' executives discuss corporate strategy 7.2 percent more than inactive firms. Thus, once again we find strong corroborating evidence in support of our predictions.

Table 3 provides the model estimation results based on the sample of M&A active firms. As before, we find support for hypotheses 1 and 2, which predict a higher use of the corporate strategy dictionary terms for M&A-active firms by both analysts and executives, respectively. More specifically, in all cases, M&A activity (whether discretized or in continuous form) is positively associated with the use of corporate keywords (i.e., $\beta = 0.86$ and $\beta = 0.26$ for analysts, and $\beta = 0.64$ and $\beta = 0.20$ for executives, all $p < 0.0001$). In economic terms, firms above the median activity level in M&A are questioned about corporate strategy 8.6 percent more compared to the firms less engaged in acquisitions. Similarly, executives from firms above the median level in M&A discuss corporate strategy 6.4 percent more compared to less active firms. Combined, these results offer additional evidence that the exchange and diffusion of information surrounding a firm's corporate strategy is affected by the firm's M&A activity levels, as predicted.

The last portion of the analysis is intended to shed light on whether the information exchange on firms' corporate strategy that unfolds as a function of their M&A activity is ultimately useful to outsiders. To answer this question, we estimate regressions that use the accuracy of analysts' predictions of firms' earnings in the quarter following the focal earnings call as the dependent variable. The key theoretical construct on the right-hand side of the model is the use of the keywords included in the corporate dictionary, in addition to several controls that may also explain the accuracy of analysts' forecasts. The results appear in Table 4.

To mirror the structure used in the previous tables, we first split the sample between M&A-active and -inactive firms, and then only consider active firms and split them between more versus less active (across the median level of activity). We find that more analyst questions on the firm's corporate strategy lead to a larger error of the firm's earnings when considering active firms (i.e., $\beta = 0.08$, $p = 0.0009$), which translates into an increase in the earnings forecast error of 8 percent in the following quarter, consistent with our prediction in H3. Interestingly, the error in the forecast is also magnified by the number of M&A transactions entered into by the firm, as shown in the models. Thus, we know that M&A activity triggers more questions from analysts just as it also leads to less precision in their earnings forecasts.

The second model of the table shows that when executives discuss their corporate strategy more, the analysts forecast error in the following quarter is considerably smaller (i.e., $\beta = -0.61$, $p < 0.0001$ and over 60 percent smaller), thereby lending support for our prediction in H4. In this case, the proclivity by insiders to inform investors of the rationale of their corporate strategy (as defined by M&A) improves the ability by analysts to forecast the firm's future performance. Again, this finding holds even though M&A generally raise the complexity of the task of predicting earnings for analysts, as also shown in the model.

The last two models in the table reveal some differences, when the sample considered is only comprised of M&A-active firms. Namely, while the information disclosures by executives still reduces analysts' forecast errors (i.e., $\beta = -0.10$, $p < 0.05$ and 10 percent smaller errors, on average), the questioning by analysts no longer increases the error (i.e., $\beta = 0.005$, n.s.). It is plausible that unlike in the previous case (i.e., the first model with active and inactive firms), the questioning by analysts on active firms' corporate strategy is not as good a proxy for how challenging their earnings forecast is going to be. This could be because the marginal benefit of the additional information sought by analysts is smaller for active firms, or alternatively, analysts following firms that are routinely engaged in M&A may be better equipped to interpret the information they receive and therefore their forecast error may be unaffected by their questioning (no matter how intense). The main takeaway from the table remains that the information diffusion on firms' corporate strategy resulting from earnings calls has a generally strong effect on the ability by analysts to forecast firm future performance.

Robustness checks. Although we have sought to make sensible decisions in the design of our empirical tests, we have also had to make some decisions that may play a role in the estimated coefficients. Most notably, we decided to track firms' M&A activities for a period of 5 years prior to the focal earnings call. It is plausible that a recency effect may be at play, such that M&A deals closer to the call will be more likely to be on the agenda and up for discussion than older deals within the time window. In this sense, using a relatively long tracking period – like we do – may weaken our results, because it may cause us to include firms that were active in M&A,

¹⁴ As before, we can provide the table for the sample that includes only M&A active firms upon request.

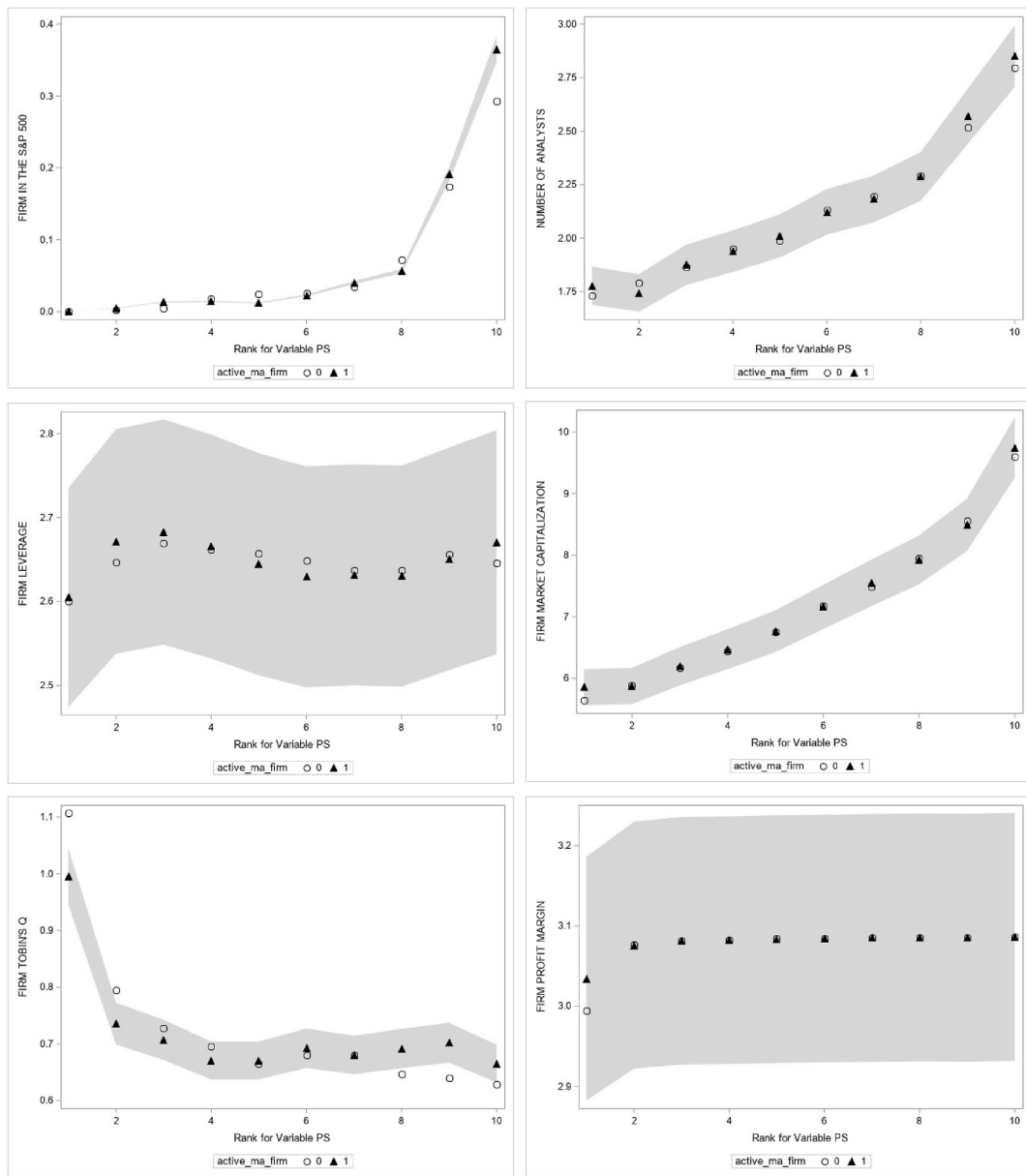


Fig. 1a. A propensity score stratification and group matching (full sample, 5 percent banding SHOWN).

but only years prior to the call. However, this approach also offers the benefit of expanding the base sample of active firms, thereby providing a more robust benchmark from which to build the matched sample.

When we change the 5-year window to 1 year, we are left with only 6,281 active firms, which are matched with 18,298 inactive ones. As expected, the more recent nature of the M&A activity experienced by firms results in far more intense scrutiny by analysts: Active firms were asked about corporate strategy 148 percent more than inactive firms (i.e., both $p < 0.0001$). For their part, executives of active firms discussed their corporate strategy 123 percent more than their counterparts. Turning to the analysts' forecast error, we find that active firms are subject to a greater error, when they are probed on corporate strategy (i.e., 14 percent larger error size, $p < 0.0001$). In contrast, the forecast error is smaller when executives discuss this topic (i.e., 44 percent smaller error size, $p < 0.0001$). Similar findings – all consistent with our predictions and main results – were found for the comparison of highly- and lowly-active firms.¹⁵

¹⁵ The results for the comparison of highly- and lowly-M&A active firms, along with additional ones using alternate time windows can be obtained from the authors on requests.

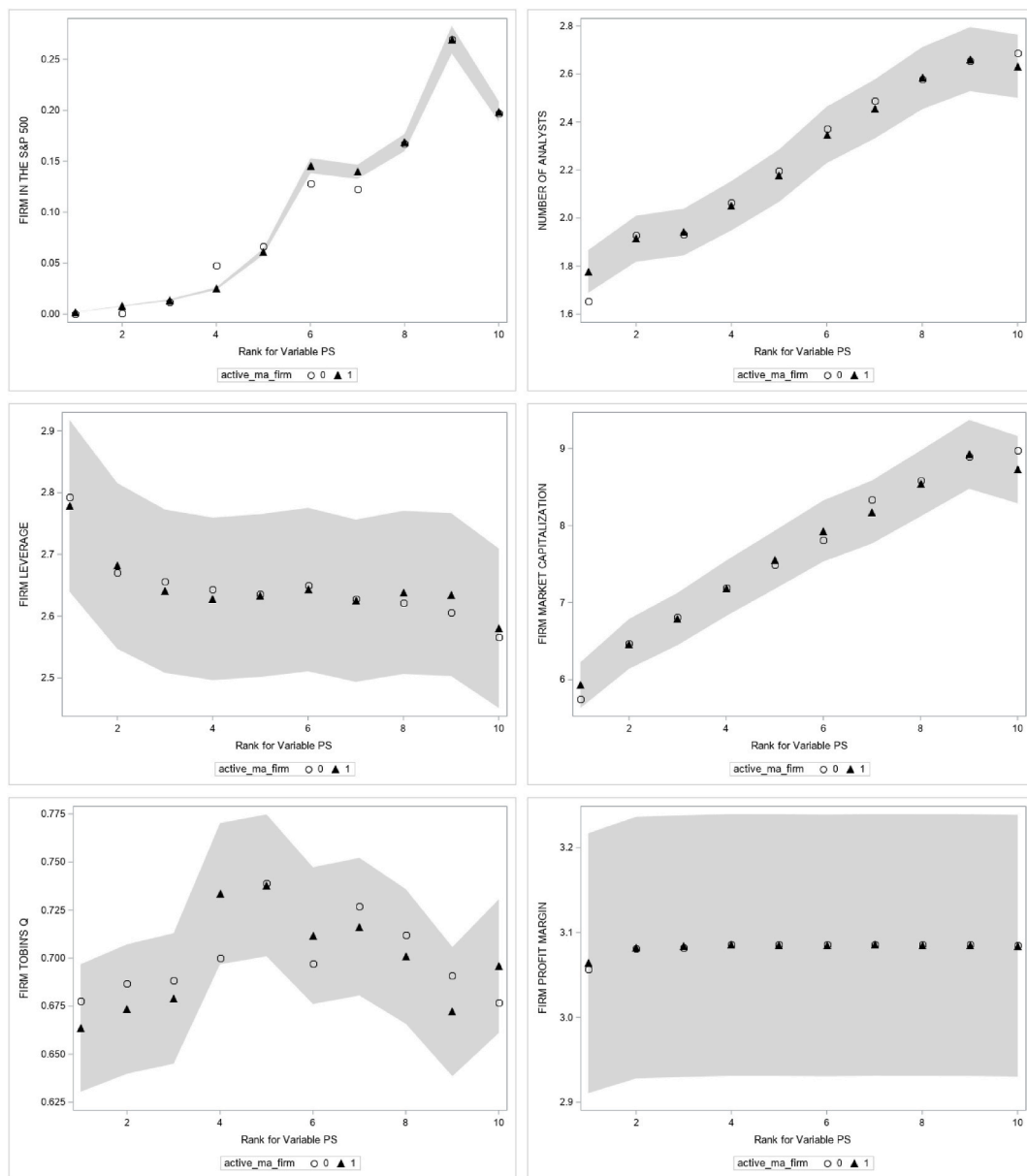


Fig. 1b. B propensity score stratification and group matching (active firms sample, 5 percent banding SHOWN).

Table 1

Descriptive statistics and correlation table for FULL matched SAMPLE^a.

Variable	Mean	SD	1	2	3	4	5	6
1. Total corporate words Analysts	1.82	2.29						
2. Total corporate words Executives	9.67	8.49	0.52					
3. Active M&A firm	0.26	0.44	0.16	0.22				
4. Firm in S&P 500	0.07	0.25	0.04	0.01	0.05			
5. Firm leverage	2.26	3.13	-0.01	0.016	0.05	0.07		
6. Firm size (\$MM)	5321	11324	0.08	0.05	0.19	0.25	0.06	
7. Firm profit margin	-0.02	0.67	0.06	0.05	0.03	0.04	0.05	0.08

^a N = 55,928. The sample is obtained by estimating propensity scores (PS) for the likelihood of an M&A deal in the 5 years preceding the focal earning call. We match active and inactive firms by PS (i.e., within 0.10 of each other), as well as by quarter, industry and whether they belong in the S&P 500 index. Pearson correlations are shown for continuous variables and Spearman for dichotomous variables. All correlation coefficients are statistically significant at $p < 0.001$.

Table 2
Tobit estimation results for matched SAMPLE (ACTIVE VS. Inactive FIRMS)^a.

	Analysts' Questions			Executives' Commentary		
	Controls only	Active vs. inactive firms	Total M&A deals	Controls only	Active vs. inactive firms	Total M&A deals
Intercept	-8.1831 1.2446 <0.0001	-7.7631 1.2282 <0.0001	-7.5701 1.2307 <0.0001	-0.6624 0.4970 0.1826	-0.4958 0.4879 0.3095	-0.2959 0.4870 0.5435
Active M&A firm		0.8785 0.0324 <0.0001			0.7183 0.0157 <0.0001	
Total M&A deals			0.2838 0.0103 <0.0001			0.2412 0.0050 <0.0001
Firm in the S&P 500 index	-0.3183 0.0601 <0.0001	-0.2873 0.0596 <0.0001	-0.3075 0.0596 <0.0001	-0.2502 0.0293 <0.0001	-0.2281 0.0288 <0.0001	-0.2423 0.0287 <0.0001
Firm leverage ^b	-0.6207 0.0718 <0.0001	-0.6273 0.0713 <0.0001	-0.6230 0.0713 <0.0001	-0.1276 0.0347 0.0002	-0.1303 0.0341 0.0001	-0.1284 0.0340 0.0002
Firm market capitalization ^b	0.2194 0.0103 <0.0001	0.1659 0.0104 <0.0001	0.1611 0.0104 <0.0001	0.0138 0.0049 0.0049	-0.0290 0.0049 <0.0001	-0.0337 0.0049 <0.0001
Firm profit margin ^b	2.9852 0.4015 <0.0001	2.9039 0.3962 <0.0001	2.8730 0.3970 <0.0001	0.9397 0.1595 <0.0001	0.9258 0.1566 <0.0001	0.8881 0.1563 <0.0001
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sigma	3.1863 0.0129 <0.0001	3.1625 0.0128 <0.0001	3.1608 0.0128 <0.0001	1.6268 0.0050 <0.0001	1.5965 0.0049 <0.0001	0.2294 0.0049 <0.0001
χ^2	1242 <0.0001	1974 <0.0001	1998 <0.0001	1720 <0.0001	3788 <0.0001	3962 <0.0001

^a N = 55,928. The sample is obtained by matching firms active in M&A with inactive firms, using propensity score and nearest-neighbor matching techniques. The models are estimated using Tobit regressions. The dependent variables in the models above represent the count of words contained in our corporate dictionary divided by the count of total spoken words. Together, these variables account for analysts' and executives' engagement in the firms' corporate strategy. Standard errors and p-values appear below each parameter estimate.

^b These variables were logged to remedy skewness.

Another potential issue is the fact that our longitudinal sample includes the years surrounding the financial crisis, which may have created uncertainty such as to influence the content of earnings calls. An examination of the year fixed effects in our model estimations reveals that the years 2007–2009 were negatively associated with the coverage of corporate strategy by both analysts and executives (i. e., all models $p < 0.001$). Interestingly, analysts appeared to ask more questions during this period (31 vs. 24 questions, specifically), although fewer questions were in fact about corporate strategy ($p < 0.001$). We suspect that this is a direct result of the fact that analysts focused more on questions pertaining to the financial and operational effects of the crisis than other topics. Likewise, executives also covered corporate strategy topics less, partly because they were not challenged with specific questions, presumably. What is somewhat surprising is that the lower coverage of corporate strategy did not stem from lower levels of M&A activity by the firms issuing the calls, as the average number of transactions leading up to the call was slightly higher in the financial crisis years, than after those years (i.e., 0.65 vs. 0.61 deals).

Independent of the above, when we reran the models excluding the financial crisis period, none of the theoretical variables in the models shown in Tables 2 and 3 were affected, as they all kept their signs and significance levels (i.e., all $p < 0.0001$). Similarly, the models shown in Table 4, which relate the difference between forecast and actual earnings in the quarter following the firm's focal earnings call, yield similar inferences for our final hypothesis. Specifically, a higher level of corporate strategy questioning by analysts in a given earnings call is associated with an increase in the forecast error of 7.0 percent for all firms ($p < 0.0001$), and of 3.2 percent for active firms (n.s.) in the following quarter. For their part, executives' more extensive use of corporate strategy words meets with a 58.4 (7.2) percent decrease in forecast error for all (active) firms ($p < 0.0001$ and $p = 0.024$, respectively). Thus, at least insofar as the relationships we are exploring, the financial crisis does not seem to play an important role in shaping the mechanisms connecting firms' M&A activity and information disclosure in earnings calls.

The next set of tests is aimed at the plausible concern that whenever executives discuss corporate strategy in earnings calls, they may do so either because they were directly asked about the topic, or they covered corporate strategy topics spontaneously. If the former scenario was the prevailing one, there might be legitimate concerns surrounding the independence of the choice of covering this topic by executives. However, just because an analyst may raise a question on corporate strategy, it does not follow that the executive will choose to cover the topic extensively, if at all. The concerns tied to opportunistic incentives and the competitive implications of "spilling the beans" on corporate strategy apply, and executives are certainly afforded discretion with respect to how they answer questions. Even still, we wished to dig into the empirical aspect of this question and went back to the data, in order to determine how often coverage of corporate strategy by an executive directly followed a question on the topic by an analyst. To answer this question,

Table 3Tobit estimation results for matched SAMPLE (ACTIVE FIRMS only)^a.

	Analysts' Questions			Executives' Commentary		
	Controls only	Above/below median	Total M&Adeals	Controls only	Above/below median	Total M&A deals
Intercept	-3.8200 2.1169 0.0711	-2.9792 2.1052 0.1570	-3.2777 2.1060 0.1196	-3.8608 1.0748 0.0003	-3.3131 1.0633 0.0018	-3.5112 1.0631 0.0010
Active M&A firm		0.8590 0.0584 <0.0001			0.6401 0.0299 <0.0001	
Total M&A deals			0.2588 0.0176 <0.0001			0.1953 0.0091 <0.0001
Firm in the S&P 500 index	-0.4126 0.0972 <0.0001	-0.2817 0.0971 0.0037	-0.3205 0.0969 0.0009	-0.0817 0.0497 0.1000	0.0113 0.0493 0.8188	-0.0159 0.0493 0.7476
Firm leverage ^b	-1.3219 0.1457 <0.0001	-1.2679 0.1450 <0.0001	-1.2591 0.1450 <0.0001	0.3851 0.0741 <0.0001	0.4210 0.0733 <0.0001	0.4276 0.0733 <0.0001
Firm market capitalization ^b	0.1332 0.0188 <0.0001	0.0352 0.0198 0.0754	0.0511 0.0195 0.0088	-0.0943 0.0095 <0.0001	-0.1653 0.0100 <0.0001	-0.1541 0.0098 <0.0001
Firm profit margin ^b	2.5810 0.6796 0.0003	2.3991 0.6757 0.0004	2.3954 0.6761 0.0004	1.9411 0.3451 <0.0001	1.8309 0.3414 <0.0001	1.8195 0.3414 <0.0001
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sigma	3.4121 0.0214 <0.0001	3.3938 0.0212 <0.0001	3.3932 0.0212 <0.0001	1.8178 0.0090 <0.0001	1.7981 0.0089 <0.0001	1.7979 0.0089 <0.0001
χ^2	568 <0.0001	784 <0.0001	784 <0.0001	952 <0.0001	1406 <0.0001	1408 <0.0001

^a N = 20,990. The sample includes only firms with at least one M&A deal. We match firms at or above the median M&A activity level with firms below the median by propensity score and nearest-neighbor matching techniques. The models are estimated using Tobit regressions. The dependent variables in the models above represent the count of words contained in our corporate dictionary divided by the count of total spoken words. Together, these variables account for analysts' and executives' engagement in the firms' corporate strategy. Standard errors and p-values appear below each parameter estimate.

^b These variables were logged to remedy skewness.

we examined the individual intervention by call participants, rather than using the earnings call as the unit of analysis. Specifically, we found that of the 2,038,415 executive responses, 179,385 touched on corporate strategy. Of these, 52,328 did so as a direct result of an analyst question, while 127,057 were instances of spontaneous coverage by executives. This translates to a ratio of nearly 2.5 of the latter to 1 of the former, and it demonstrates that executives often choose to bring up corporate strategy without a direct question about the topic. Also interestingly, the average length of an executive response that covered corporate strategy was substantially longer if the coverage was spontaneous than otherwise (i.e., $t = 14.67$, $p < 0.0001$).

Taking this supplementary analysis to the multivariate setting, and therefore returning to the individual transcript as the unit of analysis, we adopted two alternate measures to control for whether the use of corporate dictionary words by executives was partially explained by direct questioning by analysts. Namely, we first introduced an indicator variable *Analyst leading question*, which takes a value of 1 if at least once executives were asked about corporate strategy during a given earning call, and 0 otherwise. As an alternative, we also introduce a continuous variable *Analyst leading question amount* that measures the amount of corporate strategy words used by analysts during the call. We then re-estimated the models shown in Tables 2 and 3 below with the addition of either control. In all 4 models examining executives, the key theoretical variables *Active M&A firm* and *Total M&A deals* remained positive and highly consistent (i.e., all $p < 0.0001$). It is worth noting that the two controls were also positive and significant, as predicted. In sum we can conclude that, while being asked directly by analysts increases the incidence of the coverage of corporate strategy, the association between M&A activity leading up to the earnings call and the amount of executive coverage of corporate strategy remains strong independent of the presence or number of leading questions on the topic.

As a final robustness check, we considered the possibility of dropping firms operating in the financial sector, as is customarily done in the broad strategy literature. It should be noted that we do control for industry fixed effects in the models, and that we require for each matching and matched observation to be in the exact same industry. These steps should significantly alleviate the potential issue of cross-industry heterogeneity in the use of the corporate strategy dictionary words. Even still, we think it is worth exploring whether excluding such firms might produce distinct results from the ones we present in the main section of the paper. This step caused a loss of 14,044 (7,860) observations, leaving us with 41,884 (13,130) transcripts for the full (M&A active firms) sample. We find nearly identical results for the four models estimating the executives' and analysts' use of the corporate dictionary for all firms, and M&A active firms alike (i.e., for Tables 2 and 3 in the paper, all $p < 0.0001$). Turning to the analyst forecast error, again the results are very close to the original set, with the only difference being that a high use of the corporate strategy dictionary by executives of highly active

Table 4
Determinants of analysts' forecast errors^a.

	All firms (post initial matching)		All active firms	
	Analysts (N = 105,887)	Executives (N = 101,084)	Analysts (N = 27,302)	Executives (N = 25,855)
Intercept	9.6287	122.1657	7.1710	22.0113
	1.4007	1.6397	2.5900	4.3212
	<0.0001	<0.0001	0.0056	<0.0001
Corporate dictionary intensive	0.0794	-0.6055	0.0045	-0.1016
	0.0239	0.0288	0.0453	0.0492
	0.0009	<0.0001	0.9213	0.0391
Transcript length (analysts) ^b	-0.2365	0.1807	-0.2077	-0.1397
	0.0246	0.0288	0.0422	0.0485
	<0.0001	<0.0001	<0.0001	0.0039
Transcript length (executives) ^b	0.0229	-0.2726	0.0876	0.0782
	0.0088	0.0104	0.0145	0.0153
	0.0093	<0.0001	<0.0001	<0.0001
Total M&A deals	0.0311	0.0437	0.0292	0.0605
	0.0087	0.0107	0.0125	0.0143
	0.0004	<0.0001	0.0199	<0.0001
Firm in the S&P 500 index	-0.0281	0.4673	0.2601	0.4486
	0.0526	0.0733	0.0920	0.1137
	0.5936	<0.0001	0.0047	<0.0001
Number of analysts	-0.0349	0.0177	-0.0186	-0.0236
	0.0022	0.0029	0.0042	0.0047
	<0.0001	<0.0001	<0.0001	<0.0001
Firm leverage ^c	-0.5572	-1.9878	-0.0267	-0.2839
	0.0526	0.0587	0.1091	0.1111
	<0.0001	<0.0001	0.8064	0.0106
Firm market capitalization ^c	0.0440	-0.3017	-0.1354	-0.1477
	0.0103	0.0132	0.0196	0.0214
	<0.0001	<0.0001	<0.0001	<0.0001
Firm profit margin ^c	-2.3799	-36.6869	-1.8225	-6.3735
	0.4526	0.5319	0.8343	1.3985
	<0.0001	<0.0001	0.0289	<0.0001
Industry fixed effects	Yes	Yes	Yes	Yes
Model F	171.66	741.53	21.46	21.02
	<0.0001	<0.0001	<0.0001	<0.0001

^a All samples are obtained by matching firms with a high use of corporate dictionary (i.e., at or above the median) with firms with a low use of the same (i.e., below the median), using propensity score and nearest-neighbor matching techniques. The models are estimated using OLS regressions. The dependent variables in the models above represent the absolute difference between the median of the analysts' forecast of firms' earnings and their actual earnings in the quarter following the focal earnings call, divided by the median of the analyst' forecast. Standard errors and p-values appear below each parameter estimate.

^b The parameter estimates and corresponding standard errors are multiplied by 1,000 for legibility purposes.

^c These variables were logged to remedy skewness.

firms (as compared to lowly active ones, as per the last column of Table 4) is still associated with a decrease in the magnitude of the earnings forecast error, but this value falls just short of reaching statistical significance at conventional levels (i.e., $p = 0.109$).

7. Discussion

7.1. Contributions

We advance applied information economics research by considering capital market participants and the information asymmetries that they and company executives contend with in M&A. Unlike in other market contexts in which the theory has been deployed (e.g., used cars, hiring, etc.), M&A transactions not only involve buyers and sellers, but they also give rise to information asymmetry between buyers and capital market participants. The latter provide buyers with financial resources and have a vested interest in these transactions. However, capital market participants also face adverse selection, unless full information is given to them by company executives entering into M&A transactions. A wealth of studies has examined the implications of information asymmetry between buyers and sellers in M&A, and by considering the implications of information asymmetries for acquiring firms and capital market participants that have been neglected in previous research, we conclude that previous research has understated the implications of information frictions in M&A markets.

Our empirical analysis considers whether corporate communication in earnings calls reflects a firm's M&A activity. We also examine whether information disclosure reduces the effects of adverse selection otherwise borne by capital market participants. We build a corporate strategy dictionary and text-analyze the contents of millions of questions and answers in over 80,000 conference calls. We implement propensity-score matching algorithms to mitigate concerns surrounding sample selection and endogeneity, thereby isolating the implications of M&A activity for information disclosures to capital markets. The upshot of these analyses is that

we find strong empirical support for our predictions. Specifically, both analysts and company executives discuss a firm's corporate strategy more when the firm is involved in M&A activity in the period of time leading up to the call. In addition, the more M&A transactions the firm has engaged in, the more the firm's corporate strategy becomes the subject of discussion during the call. Interestingly, executives often discuss corporate strategy even when not prompted to do so by analysts, yet we see far less focus on corporate strategy topics during the financial crisis. Lastly, the accuracy of analysts' earnings forecasts decreases with aggressive questioning of the firm's corporate strategy by analysts, and it increases with time devoted by executives to explaining the firm's corporate strategy.

7.2. Research implications and future research

There are a number of implications and interesting follow-on research opportunities brought forth by this study. At the broadest theoretical level, information economics in strategy has become a central theoretical framework used to explain firms' corporate strategy agendas across a broad spectrum of corporate activities, such as mergers and acquisitions, but also strategic alliances, joint ventures, corporate venture capital, licensing, etc. While our focus is on M&A, there are clear opportunities to explore whether the complexity of firms' corporate strategy portfolios involving different transactions affect the quantity and quality of information flows from executives to capital market participants. This sort of investigation promises to be fruitful because the pursuit of competitive advantage through corporate strategy can be highly heterogeneous across firms.

Without a doubt, a great deal of research has gone into whether and how these diverse pursuits allow firms to outperform their peers, by creating market power, superior access to resources and markets, preemption of entry, etc. However, we know little about the relationship between the complexity of a firm's corporate strategy and the adverse selection problem affecting its stakeholders, where this problem is known to create a direct and consequential impact on the firm's ability to raise capital and compete in the first place. This paper lays preliminary foundations for this line of research, but future work along the lines we draw would be highly informative and beneficial to the advancement of knowledge in this area. For instance, future research might examine other types of corporate deals besides M&A (e.g., foreign direct investment, technology partnerships, etc.). It would also be valuable to extend this analysis to explore firms based in other countries subject to different information intermediaries and institutions that can shape both their corporate strategies and ability to obtain financial and other resources in factor markets (e.g., [Ahuja and Yayavaram, 2011](#)). We also think it would be interesting to examine equity market responses to previously announced M&A transactions leading up to the earnings calls, as well as the method of payment used to execute these deals, in order to determine whether such factors may trigger differential questioning on the topic of corporate strategy.

Remaining within the confines of M&A research, there are several promising avenues that deserve attention. First, while we distinguish firms based on whether or not they engage in M&A and based on how much they engage in M&A, there are opportunities to consider the qualitative characteristics of different M&A transactions. It is well known that M&A transactions are far from homogeneous, such that some deals bring together firms from like industries, markets, corporate cultures, etc., while others involve diversification into new products, geographies and corporate cultures, instead. Our objective in this project was to begin to establish a connection between M&A activity and corporate communication, but a more nuanced examination of the relationship into the micro-features of M&A activity and their effects on information disclosures and adverse selection involving capital market participants would be useful. Such work could also take up agency considerations and explicitly consider other factors affecting tradeoffs in corporate communications (e.g., appropriation and competition concerns).

Research in such directions would inform us on the degree to which particular types of transactions impose information costs on firms and capital markets, and they might also have consequences on other stakeholders and markets. Given our focus on shareholders and information intermediaries in capital markets, it would also be interesting to extend our analysis to consider other stakeholders and factor markets. For instance, research might consider how executives explain the implications of their complex M&A transactions to suppliers, partners, potential employees, etc. The degree to which adverse selection concerns in M&A spill over to these stakeholders would provide a more complete picture of the information costs of M&A and the degree to which adverse selection within M&A deals is subject to externalities across stakeholder groups. Even our study might therefore be understating the consequences of these information frictions in M&A markets.

In exploring the heterogeneity of M&A and its consequences, there are also a number of opportunities to consider how the solutions firms use to address adverse selection on M&A may or may not affect capital market participants. To begin with, the negative effects of information asymmetry may be reduced if a seller credibly conveys an information signal to a buyer ([Spence, 1973](#)). Research in strategy has studied whether the presence of signals may help to lubricate corporate transactions, generally finding support for this hypothesis. For example, prior work has shown that signals issued when firms go public facilitate their subsequent acquisitions in M&A markets ([Ragozzino and Reuer, 2007](#)), or that prior alliances with prominent partners function certify the quality of entrepreneurial firms, which in turn lends them needed credibility (e.g., [Sorenson and Stuart, 2001](#); [Chang, 2004](#); [Lee et al., 2011](#)). To our knowledge, none of this work has examined the role of signals as a tool to reduce the effects of information asymmetry between firms and their stakeholders while the firm is engaging in corporate transactions such as M&A. For instance, the very same signals on a target's resources and prospects that help firms come together in corporate transactions may also alleviate adverse selection concerns for shareholders and allow for the well-functioning of capital markets, in particular. Our research has indicated that information costs surrounding M&A might be higher than previous research has suggested, because it has not addressed the adverse selection problems that shareholders face. By contrast, the signals that firms rely upon when carrying out M&A and other corporate transactions might also be beneficial to exchanges in other factor markets (e.g., capital markets, markets for suppliers, etc.), which suggests that their benefits might also be understated in previous research. For example, within the context of our research, it would be interesting to

examine whether acquisitions in which signals are issued by a target firm result in less questioning of the transaction by analysts and in more precise earnings forecasts, as well. Therefore, we suggest that the role of signals does not stop with facilitating a corporate deal, but it potentially extends into corporate communication and the workings of capital markets. We would also note that other remedies that a firm relies upon in alleviating adverse selection in M&A (e.g., contingent payments, contractual safeguards, etc.) might not have such spillover effects on information intermediaries and capital markets, however.

Along similar lines, it would be interesting to examine whether pre-acquisition learning plays a role in corporate communication and information asymmetry between companies and their stakeholders. For example, if a company decided to integrate vertically and acquire a supplier with which it had worked for several years, then one would expect that the need to unearth information about the value-adding properties of this acquisition would be lower than if the acquisition involved an upstream firm with no prior relationship to the acquirer. Prior work has informed us of the strategic benefits of staging investments such as these, in order to reduce asymmetric information between buyers and sellers (Mitchell and Singh, 1992; Balakrishnan and Koza, 1993; Mody, 1993; Hagedoorn and Sadowski, 1999; Porri, 2004). While such successive exchanges and sequential investment approaches might also reduce information costs of shareholders and other stakeholders, we know nothing of whether this sort of learning-based approach to M&A yields similar advantages when the relationship of interest is between company insiders and the firm's shareholders and other stakeholders.

Relatedly, our findings indicate that a larger number of analysts attending a conference call tends to reduce the amount of questioning on firms' corporate activity. While a deep dive of the reasons for this relationship lies outside the scope of this work, a potentially intriguing question that stems from this result is that the quality and quantity of information disclosure issued by firms may depend on analyst attendance. If so, some firms may "get away" with less disclosure, which may then trigger different outcomes in capital markets and in the competitive environment: On the one hand, prospective shareholders and bondholders will have less information by which to evaluate the company. On the other hand, companies will be less compelled to provide information on their corporate strategies to competitors. Investigating the relationships between analyst attendance, information disclosure and firm performance would be valuable in future research.

8. Conclusion

In sum, this paper offers an initial attempt to bring to the fore the problem of adverse selection in corporate strategy, not as it pertains to the buyers and sellers directly involved in corporate deal making, but rather in the context of acquiring firms and their own shareholders. Despite the maturity of the M&A literature and corporate strategy field more generally, the implications of M&A and information asymmetries between firms and shareholders have been neglected. This likely reflects the greater attention given to moral hazard concerns as well as the direct, immediate application of theoretical models from information economics that do not involve shareholders as third parties directly in these transactions (e.g., a single buyer and seller in a transaction for a used car, insurance contract, etc.). However, the follow-on consequences of M&A for shareholders can have important implications for firms' abilities to raise external resources and ultimately compete in their marketplaces. Our theory and evidence therefore suggest that prior acquisition studies have understated the consequences of information frictions in M&A markets. Research along the lines we have suggested would help to connect M&A studies with other current streams of strategy research (e.g., on stakeholders, analysts, information disclosures, etc.), and we hope that our study encourages future research that better accounts for the implications of M&A for other factor markets.

Author statement

Roberto Ragozzino was the primary author of the paper, while Jeffrey Reuer was the second author. The authorship order reflects the amount of contribution brought forth by each author.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.lrp.2023.102393>.

References

- Affleck-Graves, J., Callahan, C.M., Chipalkatti, N., 2001. Earnings predictability, information asymmetry and market liquidity. *J. Account. Res.* 40 (3), 561–583.
- Ahuja, G., Yayavaram, S., 2011. Explaining influence rents: the case for an institutions-based view of strategy. *Organ. Sci.* 22 (6), 1631–1652.
- Akerlof, G.A., 1970. The market for 'lemons': qualitative uncertainty and the market mechanism. *Q. J. Econ.* 84 (3), 488–500.
- Angwin, D.N. (Ed.), 2007. *Mergers and Acquisitions*. Blackwell Publishing, Oxford.
- Balakrishnan, S., Koza, M.P., 1993. Information asymmetry, adverse selection, and joint ventures. *J. Econ. Behav. Organ.* 20 (1), 99–117.
- Barney, J.B., 1986. Strategic factor markets: expectations, luck, and business strategy. *Manag. Sci.* 32 (10), 1231–1241.
- Benner, M.J., Zenger, T., 2016. The lemons problem in markets for strategy. *Strat. Sci.* 1 (2), 71–89.
- Bochkay, K., Hales, J., Chava, S., 2020. Hyperbole or reality? Investor response to extreme language in earnings conference calls. *Account. Rev.* 95 (2), 31–60.

- Borochin, P., Ghosh, C., Huang, D., 2019. Target information asymmetry and takeover strategy: insights from a new perspective. *Eur. Financ. Manag.* 25 (1), 38–79.
- Bowen, R.M., Davis, A.K., Matsumoto, D.A., 2002. Do conference calls affect analysts' forecasts? *Account. Rev.* 77 (2), 285–316.
- Bushee, B.J., Matsumoto, D., Miller, G.S., 2004. Managerial and investor responses to disclosure regulation: the case of reg FD and conference calls. *Account. Rev.* 79 (3), 617–643.
- Bushee, B.J., Gow, L.D., Taylor, D.J., 2018. Linguistic complexity in firm disclosures: obfuscation or information? *J. Account. Res.* 56 (1), 85–121.
- Cain, M.D., Denis, D.J., Denis, D.K., 2011. Earnings: a study of financial contracting in acquisition agreements. *J. Account. Econ.* 51 (1–2), 151–170.
- Capron, L., Shen, J.-C., 2007. Acquisitions of private vs. public firms: private information, target selection, and acquirer returns. *Strat. Manag. J.* 28 (9), 891–911.
- Cen, L., Gilles, H., Wei, K.C.J., 2013. The role of anchoring bias in the equity market: evidence from analysts' earnings forecasts and stock returns. *J. Financ. Quant. Anal.* 48 (1), 47–76.
- Chang, S.J., 2004. Venture capital financing, strategic alliances, and the initial public offerings of Internet startups. *J. Bus. Ventur.* 19 (5), 721–741.
- Chemmanur, T.J., Fulghieri, P., 1999. A theory of the going-public decision. *Rev. Financ. Stud.* 12 (2), 249–279.
- Chemmanur, T.J., Paeglis, I., Simonyan, K., 2009. The medium of exchange in acquisitions: does the private information of both acquirer and target matter? *J. Corp. Finance* 15 (5), 523–542.
- Chen, S., Matsumoto, D.A., 2006. Favorable versus unfavorable recommendations: the impact on analyst access to management-provided information. *J. Account. Res.* 44 (4), 657–689.
- Cheng, P., Li, L., Tong, W.H.S., 2016. Target information asymmetry and acquisition price. *J. Bus. Finance Account.* 43 (7/8), 976–1016.
- Coller, M., Yohn, T.L., 1997. Management forecasts and information asymmetry: an examination of bid-ask spreads. *J. Account. Res.* 35 (2), 181–191.
- Collins, W.A., Hopwood, W.S., 1980. A multivariate analysis of annual earnings forecasts generated from quarterly forecasts of financial analysts and univariate time-series models. *J. Account. Res.* 18 (2), 390–406.
- Cuyper, I.R.P., Cuyper, Y., Martin, X., 2017. When the target may know better: effects of experience and information asymmetries on value from mergers and acquisitions. *Strat. Manag. J.* 38 (3), 609–625.
- Dalton, D.R., Hitt, M.A., Certo, S.T., Dalton, C.M. The fundamental agency problem and its mitigation. *Acad. Manag. Ann.*, 1: 1–64..
- Datar, S., Frankel, R., Wolfson, M., 2001. Earnings: the effects of adverse selection and agency costs on acquisition techniques. *J. Law Econ. Organ.* 17 (1), 201–238.
- Easterwood, J.C., Nutt, S.R., 1999. Inefficiency in analysts' earnings forecasts: systematic misreaction or systematic optimism? *J. Finance* 54 (5), 1777–1797.
- Eckbo, B.E., Giammarino, R.M., Heinkel, R.L., 1990. Asymmetric information and the medium of exchange in takeovers: theory and tests. *Rev. Financ. Stud.* 3 (4), 651–675.
- Feldman, E.R., 2020. Corporate strategy: past, present, and future. *Strateg. Manag. Rev.* 1 (1), 179–206.
- Fishman, M.J., 1989. Preemptive bidding and the role of the medium of exchange in acquisitions. *J. Finance* 44 (1), 41–57.
- Fowler, K.L., Schmidt, D., 1989. Determinants of tender offer post-acquisition financial performance. *Strat. Manag. J.* 10 (4), 339–350.
- Francis, J., Philbrick, D., 1993. Analysts' decisions as products of a multi-task environment. *J. Account. Res.* 31 (2), 216–230.
- Frankel, R., Johnson, M., Skinner, D.J., 1999. An empirical examination of conference calls as a voluntary disclosure medium. *J. Account. Res.* 37 (1), 133–150.
- Gao, H., Yu, T., Cannella Jr., A.A., 2016. The use of public language in strategy: a multidisciplinary review and research agenda. *J. Manag.* 42 (1), 21–54.
- Gavetti, G., Rivkin, J.W., 2007. On the origin of strategy: action and cognition over time. *Organ. Sci.* 18 (3), 420–439.
- Ghemawat, P., 2005. *Strategy and the Business Landscape*, second ed. Prentice Hall, Englewood Cliffs, New Jersey.
- Gilson, R.J., Schwartz, A., 2005. Understanding MACs: moral hazard in acquisitions. *J. Law Econ. Organ.* 21 (2), 330–358.
- Graebner, M.E., 2009. Caveat venditor: trust asymmetries in acquisitions of entrepreneurial firms. *Acad. Manag. J.* 52 (3), 435–472.
- Graf-Vlachy, L., Bundy, J. & Hambrick, D. C. Effects of an Advancing Tenure on CEO Cognitive Complexity. *Organization Science*, forthcoming..
- Grewal, J., Riedl, E.J., Serafeim, G., 2019. Market reaction to mandatory nonfinancial disclosure. *Manag. Sci.* 65 (7), 3061–3084.
- Grossman, S.J., Stiglitz, J.E., 1980. On the impossibility of informationally efficient markets. *Am. Econ. Rev.* 70 (3), 393–408.
- Hagedoorn, J., Sadowski, B., 1999. The transition from strategic technology alliances to mergers and acquisitions: an exploratory study. *J. Manag. Stud.* 36 (1), 87–107.
- Harris, R.D.F., Wang, P., 2013. Model-based earnings forecasts vs. financial analysts' earnings forecasts. *Br. Account. Rev.* 51 (4), 424–437.
- Harrison, J.S., Thurgood, G.R., Boivie, S., Pfarrer, M.D., 2019. Measuring CEO personality: developing, validating, and testing a linguistic tool. *Strat. Manag. J.* 40 (8), 1316–1330.
- Healy, P.M., Palepu, K.G., 1993. The effect of firms' financial disclosure strategies on stock prices. *Account. Horiz.* 7 (1), 1.
- Healy, P.M., Palepu, K.G., 2001. Information asymmetry, corporate disclosure, and the capital markets: a review of the empirical disclosure literature. *J. Account. Econ.* 31 (1–3), 405–440.
- Heinrichs, A., Park, J., Soltes, E.F., 2019. Who consumes firm disclosures? Evidence from earnings conference calls. *Account. Rev.* 94 (3), 205–231.
- Henry, E., Leone, A.J., 2005. Measuring qualitative information in capital markets research: comparison of alternative methodologies to measure disclosure tone. *Account. Rev.* 91 (1), 153–178.
- Higgins, M., Rodriguez, D., 2006. The outsourcing of R&D through acquisitions in the pharmaceutical industry. *J. Financ. Econ.* 80 (2), 351–383.
- Hollander, S., Pronk, M., Roelofs, E., 2010. Does silence speak? An empirical analysis of disclosure choices during conference calls. *J. Account. Res.* 48 (3), 531–563.
- Humphrey-Jenner, M., Sautner, Z., Suchard, J.-A., 2017. Cross-border mergers and acquisitions: the role of private equity firms. *Strat. Manag. J.* 38 (8), 1688–1700.
- Jensen, M.C., Meckling, W.H., 1976. Theory of the firm: managerial behavior, agency costs and ownership structure. *J. Financ. Econ.* 3 (4), 305–360.
- Kaplan, S.N., Weisbach, M.S., 1992. The success of acquisitions: evidence from divestitures. *J. Finance* 47 (1), 107–138.
- Kim, J.H., 2016. Market earnings expectation, measurement error in analysts' consensus forecasts and prediction of stock returns. *Account. Res.* 31 (2), 249–266.
- Kimbrough, M.D., 2005. The effect of conference calls on analyst and market underreaction to earnings announcements. *Account. Rev.* 80 (1), 189–219.
- Kimbrough, M.D., Louis, H., 2011. Voluntary disclosure to influence investors reactions to merger announcements: an examination of conference calls. *Account. Rev.* 86 (2), 637–667.
- King, D.R., Wang, G., Samimi, M., Cortes, A.F., 2021. A meta-analytic integration of acquisition performance prediction. *J. Manag. Stud.* 58 (5), 1198–1236.
- Kohers, N., Ang, J., 2000. Earnings in mergers: agreeing to disagree and agreeing to stay. *J. Bus.* 73 (3), 445–476.
- König, A., Mammen, J., Luger, J., Fehn, A., Enders, A., 2018. Silver bullet or ricochet? CEOs' use of metaphorical communication and infomediaries' evaluations. *Acad. Manag. J.* 61 (4), 1196–1230.
- Lang, M.H., Lundholm, R.J., 1996. Corporate disclosure policy and analyst behavior. *Account. Rev.* 71 (4), 467–492.
- Lee, J., 2016. Can investors detect managers' lack of spontaneity? Adherence to predetermined scripts during earnings conference calls. *Account. Rev.* 91 (1), 229–250.
- Lee, P.M., Pollock, T.G., Jin, K., 2011. The contingent value of venture capitalist reputation for entrepreneurial firms. *Strat. Organ.* 9 (1), 33–69.
- Litov, L.P., Moreton, P., Zenger, T.R., 2012. Corporate strategy, analyst coverage, and the uniqueness paradox. *Manag. Sci.* 58 (10), 1797–1815.
- Lobo, G.J., Song, M., Stanford, M.H., 2017. The effect of analyst forecasts during earnings announcements on investor responses to reported earnings. *Account. Rev.* 92 (3), 239–263.
- Löfgren, K.-G., Persson, T., Weibull, J.W., 2002. Markets with asymmetric information: the contributions of George Akerlof, Michael Spence and Joseph Stiglitz. *Scand. J. Econ.* 104 (2), 192–211.
- Macias, A., Moeller, T., 2016. Target signaling with material adverse change clauses in merger agreements. *J. Empir. Finance* 39 (A), 69–92.
- Martin, X., Shalev, R., 2017. Target firm-specific information and acquisition efficiency. *Manag. Sci.* 63 (3), 672–690.
- Matsumoto, D., Pronk, M., Roelofs, E., 2011. What makes conference calls useful? The information content of managers' presentations and analysts' discussion sessions. *Account. Rev.* 86 (4), 1383–1414.
- Mayew, W.J., 2008. Evidence of management discrimination among analysts during earnings conference calls. *J. Account. Res.* 46 (3), 627–659.
- Mayew, W.J., Venkatachalam, M., 2012. The power of voice: managerial affective states and future firm performance. *J. Finance* 67 (1), 1–43.

- Mayew, W.J., Sharp, N.Y., Venkatachalam, M., 2013. Using earnings conference calls to identify analysts with superior private information. *Rev. Account. Stud.* 18 (2), 386–413.
- McCann, B., Reuer, J.J., Lahiri, N., 2016. Agglomeration and the choice between acquisitions and alliances: an information economics perspective. *Strat. Manag. J.* 37 (6), 1085–1106.
- Milgrom, P., Stokey, N., 1982. Information trade and common knowledge. *J. Econ. Theor.* 26 (1), 17–27.
- Mintzberg, H., 1987. The strategy concept I: 5 ps for strategy. *Calif. Manag. Rev.* 30 (1), 11–24.
- Mitchell, W., Singh, K., 1992. Incumbents' use of pre-entry alliances before expansion into new technical subfields of an industry. *J. Econ. Behav. Organ.* 18 (3), 347–372.
- Mody, A., 1993. Learning through alliances. *J. Econ. Behav. Organ.* 20 (2), 151–170.
- Moeller, S.B., Schlingemann, F.P., Stulz, R.M., 2007. How do diversity of opinion and information asymmetry affect acquirer returns? *Rev. Financ. Stud.* 20 (6), 2047–2078.
- Montgomery, C.A., Wernerfelt, B., 1988. Diversification, Ricardian rents, and Tobin's q. *Rand J. Econ.* 19 (4), 623–632.
- Myers, S.C., Majluf, N.S., 1984. Corporate financing and investment decisions when firms have information that investors do not have. *J. Financ. Econ.* 13 (2), 187–221.
- National Investor Relations Institute (NIRI), 2014. Earnings Call Practices Survey. <https://www.niri.org/>.
- Officer, M.S., Poulsen, A.B., Stegemoller, M., 2009. Target-firm information asymmetry and acquirer returns. *Rev. Finance* 13 (3), 467–493.
- Pan, L., McNamara, G., Lee, J.J., Halebian, J., Devers, C.E., 2018. Give it to us straight (most of the time): top managers' use of concrete language and its effect on investor reactions. *Strat. Manag. J.* 39 (8), 2204–2225.
- Porrini, P., 2004. Can a previous alliance between an acquirer and a target affect acquisition performance? *J. Manag.* 30 (4), 545–562.
- Rabier, M.R., 2017. Acquisition motives and the distribution of acquisition performance. *Strat. Manag. J.* 38 (13), 2666–2681.
- Ragozzino, R., Reuer, J.J., 2007. Initial public offerings and the acquisition of entrepreneurial firms. *Strat. Organ.* 5 (2), 155–176.
- Ravenscraft, D.J., Scherer, F.M., 1987. Life after takeover. *J. Ind. Econ.* 36 (2), 147–156.
- Reuer, J.J., 2009. Organizational economics insights from acquisitions research. *Adv. Strat. Manag.* 26, 241–265.
- Reuer, J.J., Ragozzino, R., 2012. The choice between joint ventures and acquisitions: insights from signaling theory. *Organ. Sci.* 23 (4), 1175–1190.
- Riley, J.C., 2001. Silver signals: twenty-five years of screening and signaling. *J. Econ. Lit.* 39 (2), 432–478.
- Rothschild, M., Stiglitz, J., 1976. Equilibrium in competitive insurance markets: an essay on the economics of imperfect information. *Q. J. Econ.* 90 (4), 629–649.
- Schildt, H.A., Laamanen, T., 2006. Who buys whom: information environments and organization boundary spanning through acquisitions. *Strat. Organ.* 4 (2), 111–133.
- Shapiro, C., Stiglitz, J., 1984. Equilibrium unemployment as a worker discipline device. *Am. Econ. Rev.* 74 (3), 433–444.
- Shiller, R.J., 1989. *Market Volatility*. MIT Press, Cambridge, MA.
- Shleifer, A., 1986. Do demand curves for stocks slope down? *J. Finance* 41 (3), 579–590.
- Sorenson, O., Stuart, T.E., 2001. Syndication networks and the spatial distribution of venture capital investments. *Am. J. Sociol.* 106 (6), 1546–1588.
- Spence, M.A., 1973. Job market signaling. *Q. J. Econ.* 87 (3), 355–374.
- Stiglitz, J.E., 1973. Approaches to the economics of discrimination. *Am. Econ. Rev.* 63 (2), 287–295.
- Stiglitz, J.E., Weiss, A., 1981. Credit rationing in markets with imperfect information. *Am. Econ. Rev.* 71 (3), 393–410.
- Stiglitz, J.E., 2002. Information and the change in the paradigm in economics. *Am. Econ. Rev.* 92 (3), 460–501.
- Vanhaverbeke, W., Duysters, G., Noorderhaven, N., 2002. External technology sourcing through alliances or acquisitions: an analysis of the application-specific integrated circuits industry. *Organ. Sci.* 13 (6), 714–733.
- Veldkamp, L., 2006. Media frenzies in the market for financial information. *Am. Econ. Rev.* 96 (3), 577–601.
- Villalonga, B., McGahan, A.M., 2005. The Choice among acquisitions, alliances, and divestitures. *Strat. Manag. J.* 26 (13), 1183–1208.
- Whittington, R., Cailluet, L., Yakis-Douglas, B., 2011. Opening strategy: evolution of a precarious profession. *Br. J. Manag.* 22 (3), 531–544.
- Whittington, R., Yakis-Douglas, B., Ahn, K., 2016. Cheap talk? Strategy presentations as a form of chief executive officer impression management. *Strat. Manag. J.* 37 (12), 2413–2424.
- Yakis-Douglas, B., Angwin, D., Ahn, K., Meadows, M., 2017. Opening M&A strategy to investors: predictors and outcomes of transparency during organisational transition. *Long. Range Plan.* 50 (3), 411–422.
- Zaheer, A., Hernandez, E., Banerjee, S., 2010. Prior alliances with targets and acquisition performance in knowledge-intensive industries. *Organ. Sci.* 21 (5), 1072–1091.
- Zenger, T.R., Felin, T., Bigelow, L., 2011. Theories of the firm-market boundary. *Acad. Manag. Ann.* 5 (1), 89–133.

Dr. Ragozzino is a full Professor of Strategy and Entrepreneurship at NOVA School of Business and Economics in Lisbon, Portugal. He holds a PhD in Business Policy and Strategy from the Max M. Fisher College of Business, Ohio State University.

Before joining NOVA he was an endowed chair professor at the University of Tennessee, in Knoxville, USA. His research is in the area of mergers and acquisitions and strategic alliances with an emphasis on how these events apply to entrepreneurial ventures. Dr. Ragozzino has published in such journals as *Journal of Management*, *Journal of International Business Studies*, *Long Range Planning*, *Organization Science*, *Strategic Management Journal* and several others. He is also a member of the Strategic Management Society, the Academy of Management and the Academy of International Business. He is on the editorial boards of *Strategic Management Journal* and *Global Strategy Journal*. Dr. Ragozzino has extensive experience teaching Strategy at all levels and he also routinely participates in executive programs in the United States, Mexico and across Europe.